

Usama Ahmed

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EXPERIENCE

- University of Arizona** Tucson, AZ
Research Assistant January 2025 - present
 - Framed complex research questions in **mathematical terms** and developed **ML solutions** to study logical reasoning in LLMs; applied **supervised** training over synthetic corpora to improve model quality and generalizability.
 - Designed and executed **6 model development** experiments end-to-end – **data collection, exploratory data analysis, corpus synthesis (30K training examples), model training, testing, and selection** – applying **statistical hypothesis testing** and ablation studies to validate results.
 - Generated **structured insights** from modelling exercises and reported results with reproducible **evaluation pipelines**; communicated findings in appropriate formats for both **technical and non-technical** audiences.
- Pi Labs, Inc.** San Francisco, CA
GenAI Intern October 2025 – December 2025
 - Developed **ML-powered semantic search** and question answering pipelines integrating **BM25, dense retrieval**, and query fan-out mechanisms; applied **data mining** techniques to enhance document recall, precision, and ranking relevance.
 - Engineered **synthetic data generation workflows** using LLMs (**GPT-4o, Claude 3.5**) to create **2.17M** diverse query–document pairs; benchmarked **semantic vs. lexical retrieval** against the full **MS MARCO dataset (8.8M passages)** to drive product search quality improvements.
 - Designed **offline evaluation frameworks**, defining metrics (**F1, recall@k**) and automating gold-label generation to produce **decision-ready insights** on system performance and regressions.
 - Communicated tradeoff analyses (quality, latency, cost) and findings to **non-technical** product stakeholders as clear written narratives, directly influencing **pipeline architecture decisions**.
- University of Arizona** Tucson, AZ
Graduate Teaching Assistant – SQL/NoSQL Databases January 2025 – May 2025
 - Assisted **50+ students** in mastering **SQL** and **NoSQL** databases, covering indexing, **query optimization**, and **data modeling**; led **10+ recitation sessions** and **20+ office hours** troubleshooting complex queries and database design challenges.
 - Graded and provided feedback on **100+ assignments** evaluating **SQL queries, NoSQL schema designs, and ETL processes**; mentored students on data consistency models, **CAP theorem**, and scalability strategies.
- Afiniti Software Solutions** Washington, D.C.
Data Scientist II – Applied AI November 2022 – December 2023
 - Applied **predictive modeling** and **machine learning** techniques on large-scale transactional data (**3M+ call volume**) for major **telecom enterprise clients** (including Bouygues Telecom and Telefónica), driving a **\$5M/month** uplift in customer retention.
 - Executed the full **ML lifecycle** – problem framing, **exploratory data analysis**, feature engineering (**20+ new features** on top of **80+ existing**), **model training, testing, and deployment** – automating **Python** and **SQL** pipelines for **400K+** entities.
 - Performed **statistical analyses** and **incrementality measurement** against baseline/control policies; defined problems in **quantitative terms** to identify model gaps, improving incremental lift from **17 to 33 cents per dollar**.
 - Designed **real-time monitoring** for model health and business outcome regressions (drift, KPIs, anomalies) using **PostgreSQL, Python, and R**; delivered **executive-facing dashboards** presenting technical results to **non-technical** senior leadership.
- CarFirst** Lahore, PK
Data Science Product Manager November 2021 - August 2022
 - Built a **probabilistic classification** system using **Bayesian statistics** to support operational decisioning, boosting **auction success rate by 20%** and predicting a **3% revenue increase**.
 - Performed **ad-hoc statistical analyses** and **data mining** on operational data; translated findings into **business insights** to drive prioritization and automation of the procurement workflow.
 - Delivered recurring **data visualization** and executive-ready reporting to senior leadership, informing prioritization decisions and preventing a **PKR 40 million** loss through data-driven harm mitigation.

PROJECTS

- **Multi-Label Sentiment Classification (Neural Network – Competition)** GitHub
TensorFlow, Keras, RoBERTa, HuggingFace Transformers, Bidirectional LSTM
 - Built a **Bidirectional LSTM** neural network (**Keras, TensorFlow**) over **RoBERTa** embeddings for multi-label **sentiment classification**; architecture: Input → Bi-LSTM → GlobalMaxPooling → Dense (L2) → BatchNorm → Dropout → Sigmoid.
 - Implemented a custom **weighted binary cross-entropy loss** with per-label class balancing; optimized with **F1-score early stopping** and **ExponentialDecay** learning rate scheduling.
 - Applied **supervised learning** end-to-end: **data collection**, tokenization, **exploratory data analysis**, **model training, testing, and selection** with **statistical** benchmark-grade evaluation.
- **Algorithmic Trading System (Signal Generation + Backtesting)** GitHub
Python, yfinance, MACD, RSI, Parquet, Hugging Face Datasets, asyncio, multiprocessing
 - Built **trading bots** generating **buy/sell signals** using technical indicators (**MACD, RSI**) with **backtesting** across equities, crypto, ETFs, FX, futures, and volatility instruments; derived **business insights** from quantitative results to inform decisions.
 - Implemented a robust **OHLCV ingestion pipeline** working with **large and complex datasets** spanning 15–20 year market windows; built async multiprocessing pipeline for large-scale parallel scanning.
 - Published datasets to **Hugging Face Hub** with structured **JSON** and **CSV** exports and **data visualization** artifacts (PNG plots) for downstream consumption and reporting.
- **Synthetic Query Generation Pipeline (RAG / Search Eval)** GitHub
LangChain, BERTopic, Sentence-Transformers, BM25, llama.cpp
 - Developed a **topic-aware synthetic data generation** pipeline over Wikipedia using **BERTopic** + local **LLaMA (llama.cpp)** to create evaluation datasets for **semantic search** quality measurement.
 - Served generation workloads with **multi-GPU distributed inference**; filtered **1M+ queries** using hybrid heuristics: **BM25**, Jaccard, Sentence-Transformers, TF-IDF – producing **decision-ready** benchmark data.
- **Weak Labels Pipeline (Auto-Qrels + Triples)** GitHub
PyTorch, BM25S, BGE, Cross-Encoder, vLLM, SLURM
 - Built a **weak supervision** workflow generating training labels via **BM25S** → **BGE** dense rerank → cross-encoder → **vLLM** judge; operationalized distributed inference on **SLURM** at **million-scale** throughput.
 - Cut weak-label judging latency from **13s/query** to **~2s/query** by optimizing **vLLM** batching; established **evaluation frameworks** with **performance measurement** benchmarks for detection system quality.

EDUCATION

- **University of Arizona** Tucson, US
Master of Science - Data Science *January 2024 - May 2025*
Awards: Presidential Scholar, Dean's Honor List 2025
- **Lahore University of Management Sciences** Lahore, Pakistan
Bachelor of Science - Economics and Mathematics *August 2017 - January 2022*
Awards: Dean's Honor List 2021

SKILLS SUMMARY

- **Languages:** Python, SQL, R
- **Machine Learning:** Predictive Modeling, Supervised & Unsupervised Learning, Statistical Hypothesis Testing, Experiment Design, A/B Testing, Anomaly Detection, XGBoost, Scikit-Learn
- **ML Lifecycle:** Problem Framing, Data Collection, Exploratory Data Analysis, Model Development, Deployment, Performance Measurement
- **Deep Learning / NLP:** TensorFlow, Keras, PyTorch, HuggingFace Transformers, RAG, Semantic Search, LLM Fine-Tuning, Synthetic Data Generation, Weak Supervision
- **Data & Analytics:** PostgreSQL, MySQL, MongoDB, Parquet, JSONL, Data Mining, KPI Reporting, Forecasting
- **Visualization & Reporting:** Matplotlib, Seaborn, Plotly, Power BI, ggplot2, R Markdown, Shiny, Executive Dashboards
- **Infra & Tooling:** Git/GitHub, Bash, asyncio, multiprocessing, SLURM (HPC), distributed inference (vLLM, multi-GPU)